

EDUCATION

Children's Hospital of Philadelphia	Philadelphia, Pennsylvania
Postdoctoral Fellow Pediatric Cardiology	2021–2023
Cornell University	Ithaca, New York
Ph.D., Structural Mechanics Minor: Computational Science and Engineering	May 2021
M.S., Structural Engineering	August 2018
B.S., Civil Engineering <i>Magna Cum Laude</i>	May 2015

ACADEMIC POSITIONS

University of Pennsylvania	Philadelphia, Pennsylvania
Mechanical Engineering and Applied Mechanics <i>Research Assistant Professor</i>	2025–Present
Children's Hospital of Philadelphia	Philadelphia, Pennsylvania
Department of Pediatric Cardiology <i>Research Assistant Professor</i>	2025–Present
Department of Pediatric Cardiology <i>Research Associate Scientist</i>	2023–2025
Department of Anesthesiology and Critical Care Medicine <i>Research Associate Scientist</i>	2023–2025

SELECTED AWARDS, GRANTS, AND HONORS

Additional Ventures Single Ventricle Research Fund	2025–2028
National Institute of Health (NIH) K25 Mentored Career Development Award	2023–2027
NIH National Research Service Award T32 Institutional Training Fellowship	2022–2023
Cornell University Ve-Sing and Tseng So Koo Award	2015
National Science Foundation sponsored Research Experiences for Undergraduates program: Best Overall Project (Ranked 1st of 8 groups)	2013
Tau Beta Pi National Engineering Honor Society	2013–Present
Chi Epsilon National Civil Engineering Honor Society	2013–Present

PREPRINTS AND ONGOING WORK

5. N. Parvez, P. K. Purohit, **W. Wu**. "An Octahedral Fibrous Constitutive Model for Heart Valve Mechanics and Function," [arXiv.org/pdf/2606.04234](https://arxiv.org/pdf/2606.04234). *Under review*.
4. S. J. Koohy, **W. Wu**, M. A. Jolley, P. Perdikaris. "Real-time Generalizable Heart Valve Mechanics for Clinical Disease Assessment via a Physics-Conditioned Neural Operator," *Under review*.
3. T. Lobo, **W. Wu**, H. I. Litt, M. Freas, V. Ferrari, J. E. Bavaria, N. D. Desai, A. M. Pouch. "Validating Image-Derived Ascending Aortic Strain: Accuracy, Robustness, and Failure Modes," *Under internal review*.
2. J. Maheshwaria, **W. Wu**, C. N. Zelonis, S. A. Maas, K. Sunderlande, B. Yuval, S. Ching, P. M. Sabin, A. Lasso, M. J. Gillespie, J. A. Weiss, M. A. Jolley. "Effect of Right Ventricular Outflow Tract Material Properties on Simulated Transcatheter Pulmonary Placement," arxiv.org/pdf/2601.05410. *In revision*.
1. C. N. Zelonis, J. Maheshwaria, **W. Wu**, S. A. Maas, A. Seda, K. Sunderlande, S. Ching, A. Koluda, B. Yuval, N. R. Mangine, P. M. Sabin, A. Lasso, D. W. Laurence, C. Herz, M. J. Gillespie, J. A. Weiss, M. A. Jolley. "Integrated Open-Source Framework for Simulation of Transcatheter Pulmonary Valves in Native Right Ventricular Outflow Tracts," arxiv.org/pdf/2507.06337v2. *Under review*.

REFEREED JOURNAL PUBLICATIONS

13. N. R. Mangine, D. W. Laurence, P. M. Sabin, **W. Wu**, C. Herz, C. N. Zelonis, J. S. Unger, C. Pinter, A. Lasso, S. A. Maas, J. A. Weiss, M. A. Jolley. "Effect of Parametric Variation of Chordae Tendineae Structure on Simulated Atrioventricular Valve Closure," *Annals of Biomedical Engineering*, In press, 2026.
12. **W. Wu**, M. Daemer, J. A. Weiss, A. M. Pouch, M. A. Jolley. "A Geometric Feature Tracking Approach for Noninvasive Patient Specific Estimation of Leaflet Strain from 3D Images of Heart Valves," *Annals of Biomedical Engineering*, In press, 2026.
11. R. T. Black, S. A. Maas, **W. Wu**, J. Maheshwaria, K. Tzanio, J. A. Weiss, M. A. Jolley. "An Open-Source Computational Framework for Immersed Fluid-Structure Interaction Modeling using FEBio and MFEM," *Engineering with Computers*, In press, 2026.
10. M. Nakhaei, A. M. Pouch, S. Amin, M. Daemer, C. Herz, N. Yushkevich, L. Al. Ghofaily, N. D. Desai, J. E. Bavaria, M. A. Jolley, **W. Wu**. "Biomechanically Informed Image Registration for Patient-Specific Aortic Valve Strain Analysis," *Annals of Biomedical Engineering*, In press, 2026.
9. S. Aslan, N. R. Mangine, D. W. Laurence, P. M. Sabin, **W. Wu**, C. Herz, J. S. Unger, S. A. Maas, M. J. Gillespie, J. A. Weiss, M. A. Jolley. "Simulation of Transcatheter Therapies for Atrioventricular Valve Regurgitation in an Open-Source Finite Element Simulation Framework," *Annals of Biomedical Engineering*, In press, 2026.
8. **W. Wu**, M. Daneker, C. Herz, H. Dewey, J. A. Weiss, A. M. Pouch, L. Lu, M. A. Jolley. "A Noninvasive Method for Determining Elastic Parameters of Valve Tissue Using Physics-Informed Neural Networks," *Acta Biomaterialia*, Vol. 200, 283-298, 2025.
7. **W. Wu**, M. Daneker, K. T. Turner, M. A. Jolley, L. Lu. "Identifying Heterogeneous Micromechanical Properties of Biological Tissues via Physics-Informed Neural Networks," *Small Methods*, Vol. 9, 2400620, 2025.
6. **W. Wu**, M. Daneker, M. A. Jolley, K. T. Turner, L. Lu. "Effective Data Sampling Strategies and Boundary Condition Constraints of Physics-Informed Neural Networks for Identifying Material Properties in Solid Mechanics," *Applied Mathematics and Mechanics*, Vol. 44, 1039-1068, 2023.
5. **W. Wu**, S. Ching, P. M. Sabin, D. W. Laurence, S. A. Maas, A. Lasso, J. A. Weiss, M. A. Jolley. "The Effects of Leaflet Material Properties on the Simulated Function of Regurgitant Mitral Valves," *Journal of the Mechanical Behavior of Biomedical Materials*, Vol. 142, 105858, 2023.
4. **W. Wu**, S. Ching, S. A. Maas, A. Lasso, P. M. Sabin, J. A. Weiss, M. A. Jolley. "A Computational Framework for Atrioventricular Valve Modeling using Open-Source Software," *Journal of Biomechanical Engineering*, Vol. 144, 101012, 2022.
3. **W. Wu**, C. J. Earls. "A New Engineering Theory Describing Oblique Free Surface Impact by Flexible Plates," *Ocean Engineering*, Vol. 256, 111473, 2022.
2. **W. Wu**, C. Bonneville, C. J. Earls. "A Principled Approach to Design using High Fidelity Fluid-Structure Interaction Simulations," *Finite Element in Analysis & Design*, Vol. 194, 103562, 2021.
1. **W. Wu**, J. W. Kosianka, H. M. Reed, C. J. Stull, and C. J. Earls. "CU-BENs: A Structural Finite Element Library," *SoftwareX*, Vol. 11, 100485, 2020.

REFEREED CONFERENCE PROCEEDINGS

2. **W. Wu**, Y. Wu, A. M. Sulentic, J. C. Gee, A. M. Pouch, M. A. Jolley. "Physics in the Loop: Integrating Biomechanics-Derived Training Data into a Neural Ordinary Differential Equation-Based Deformable Registration Framework," *Medical Imaging with Deep Learning*, 2024.
1. P. J. Hughes, W. Scott, **W. Wu**, R. J. Kuether, M. S. Allen, and P. Tiso. "Interface Reduction on Hurty/Craig-Bampton Substructures with Frictionless Contact," *Nonlinear Dynamics*, Vol. 1, Conference Proceedings of the Society for Experimental Mechanics Series, 2019.

ACTIVE RESEARCH GRANTS

A Valveless Fontan Assist Device for Patients with Single Ventricle Congenital Heart Disease

- **Funding Mechanism:** Additional Ventures Device Innovation Award
- **Period of Support:** September 1, 2026 to August 31, 2029
- **Level of Support:** \$399,828 Y1 (\$14,593 to Wu)
- **Role:** Leadership Team (**PI:** Benjamin Kozyak)

Cardiac Neural Operator: A Mechanics-Informed Engine for Cardiac Valve Analysis

- **Funding Mechanism:** Cardiac Center Innovation Award at Children's Hospital of Philadelphia
- **Period of Support:** March 30, 2026 to March 30, 2028
- **Level of Support:** \$200,000
- **Role:** Principal Investigator (**co-PI:** Paris Perdikaris)

Understanding Biomechanical Mechanisms of Atrioventricular Valve Failure in Single Ventricle Patients

- **Funding Mechanism:** Additional Ventures Single Ventricle Research Funds
- **Period of Support:** July 1, 2025 to June 30, 2028
- **Level of Support:** \$659,937
- **Role:** Principal Investigator

Development of Machine Learning Models for Cardiovascular Simulations

- **Funding Mechanism:** National Science Foundation, ACCESS
- **Period of Support:** June 20, 2025 to June 19, 2026
- **Level of Support:** \$246,100 worth of GPU hours
- **Role:** Principal Investigator

Toward Patient-Specific Computational Modeling of Tricuspid Valve Repair in Hypoplastic Left Heart Syndrome

- **Funding Mechanism:** NIH / NHLBI K25 Mentored Quantitative Research Career Development Award
- **Period of Support:** September 1, 2023 to August 31, 2027
- **Level of Support:** \$579,962
- **Role:** Principal Investigator

PAST RESEARCH GRANTS

A Novel, Non-invasive Computational Approach for Determining the Etiology of Tricuspid Regurgitation in Patients With Hypoplastic Left Heart Syndrome

- **Funding Mechanism:** NIH / NHLBI NRSA T32 Institutional Research Training Fellowship
- **Period of Support:** July 1, 2022 to August 31, 2023
- **Level of Support:** \$76,140 to Wu
- **Role:** Principal Investigator (**Program Director:** Robert J. Levy)

Deep Learning and Physics-Informed Neural Networks to Advance Single Ventricle Atrioventricular Valve Modeling

- **Funding Mechanism:** Additional Ventures Expansion Award
- **Period of Support:** July 1, 2022 to June 30, 2023
- **Level of Support:** \$50,000 for research equipment
- **Role:** Co-Investigator (**MPI:** Matthew A. Jolley, Alison M. Pouch, and Lu Lu)

Partitioned Approach, Implicit Fluid-Structure Interaction for the Study of Hydroelastic Effects in High Speed Watercraft

- **Funding Mechanism:** National Science Foundation, XSEDE
- **Period of Support:** April 20, 2020 to April 19, 2021
- **Level of Support:** \$1,157 worth of CPU hours
- **Role:** Primary Investigator (**PI:** Christopher J. Earls)

INVITED TALKS

3. **W. Wu.** "Physics-Informed Neural Networks for Solving Forward and Inverse Problems in Solid Mechanics," *National Yang Ming Chiao Tung University*, Taipei, Taiwan. May 2026.
2. **W. Wu.** "From Imaging to Mechanics: Patient-Specific Modeling of Cardiac Valves," *The University of Massachusetts Amherst*, Amherst, MA. March 2026.
1. **W. Wu.** "Determining Heterogeneous Elastic Properties of Soft Materials using Physics-Informed Neural Networks," *4th International Workshops on Advances in Computational Mechanics*, Kitakyushu, Japan. September 2024.

CONFERENCE PRESENTATIONS

25. S. J. Koohy, **W. Wu**, P. Perdikaris. "Data-Driven Modeling of Heart Valve Biomechanics Using Transformer-Based Neural Operators," *SIAM Conference on the Life Sciences*, Cleveland, OH. July 2026. [Oral].
24. T. Lobo, **W. Wu**, H. I. Litt, M. Freas, V. Ferrari, J. E. Bavaria, N. D. Desai, A. M. Pouch. "Validating Image-Derived Ascending Aortic Strain: Accuracy, Robustness, and Failure Modes," *Pendergrass Symposium at the University of Pennsylvania*, Philadelphia, PA. May 2026. [Poster].
23. R. Mallela, **W. Wu**. "Towards Explainable and Real-Time Hemodynamic Modeling of Intracranial Aneurysms Using POD-GNNs," *2026 National Premier Undergraduate Research Conference at Columbia University*, New York, NY. April 2026. [Poster].
22. R. Mallela, **W. Wu**. "Towards Explainable and Real-Time Hemodynamic Modeling of Intracranial Aneurysms Using POD-GNNs," *Penn AI in Medicine Symposium at the University of Pennsylvania*, Philadelphia, PA. April 2026. [Poster].
21. M. Nakhaei, S. Amin, C. Herz, N. Yushkevich, L. Al. Ghofaily, N. D. Desai, J. E. Bavaria, M. A. Jolley, A. M. Pouch, **W. Wu**. "Simulation Guided Aortic Valve Tracking and Strain Analysis in 4D Echocardiography," *2025 American Heart Association Scientific Sessions*, New Orleans, LA. November 2025. [Oral].
20. T. Lobo, **W. Wu**, H. I. Litt, M. Freas, S. Goldfinger, V. Ferrari, J. E. Bavaria, N.D. Desai, A. M. Pouch. "Patient-Specific Ascending Aortic Wall Shear Stress and Strain Analysis from 4D CT," *2025 American Heart Association Scientific Sessions*, New Orleans, LA. November 2025. [Poster].
19. **W. Wu**. Meet the Faculty Candidate Forum. *BMES: Biomedical Engineering Society Annual Meeting*, San Diego, CA, October 2025. [Poster].
18. S. Patel, G. Unger, B. Kozyak, **W. Wu**, C. Sung, J. R. Raney. "Bio-Inspired Valveless Pumping for Fontan Circulation: Experimental and Modeling Insights," *BMES: Biomedical Engineering Society Annual Meeting*, San Diego, CA, October 2025. [Poster].
17. **W. Wu**, M. Daneker, C. Herz, H. Dewey, J. A. Weiss, A. M. Pouch, L. Lu, M. A. Jolley. "A Noninvasive Method for Determining Elastic Properties of Valve Tissue," *The Third Joint SIAM/CAIMS Annual Meetings (AN25)*, Montreal, Canada. July 2025. [Oral].
16. **W. Wu**, M. Daneker, C. Herz, H. Dewey, J. A. Weiss, A. M. Pouch, L. Lu, M. A. Jolley. "A Noninvasive Method for Determining Elastic Properties of Valve Tissue," *18th U.S. National Congress on Computational Mechanics*, Chicago, Illinois, US. July 2025. [Oral].

15. J. S. Unger, D. W. Laurence, N. R. Mangine, **W. Wu**, S. A. Maas, J. A. Weiss, M. A. Jolley. "Gradient-Based Optimization of Chord Structure Topologies to Enhance Image-Derived Simulation Fidelity to In Vivo Heart Valve Function," 18th U.S. National Congress on Computational Mechanics, Chicago, Illinois, US. July 2025. [Oral].
14. **W. Wu**, M. Daneker, K. T. Turner, M. A. Jolley, L. Lu. "Determining Heterogeneous Elastic Properties of Soft Materials using Physics-Informed Neural Networks," 2024 Materials Science & Technology, Pittsburgh, Pennsylvania, US. October 2024. [Oral].
13. **W. Wu**, M. Daneker, K. T. Turner, M. A. Jolley, L. Lu. "An Accurate Physics-Informed Neural Network Architecture for Determining the Heterogeneous Micromechanical Elastic Properties of Biological Materials," 16th World Congress on Computational Mechanics, Vancouver, British Columbia, Canada. July 2024. [Oral].
12. **W. Wu**, Y. Wu, A. M. Sulentic, J. C. Gee, A. M. Pouch, M. A. Jolley. "Physics in the Loop: Integrating Biomechanics-Derived Training Data into a Neural Ordinary Differential Equation-Based Deformable Registration Framework", *Medical Imaging with Deep Learning*, Paris, France. July 2024. [Poster].
11. N. Mangine, P. M. Sabin, D. W. Laurence, **W. Wu**, C. Herz, C. N. Zelonis, C. Pinter, A. Lasso, S. Ching, S. A. Maas, J. A. Weiss, M. A. Jolley. "A Parametric Analysis of Chordae Tendineae Density and Branching in Finite Element Simulations of Mitral Valve Closure," 2024 Summer Biomechanics, Bioengineering and Biotransport Conference, Lake Geneva, Wisconsin, US. June 2024. [Oral].
10. C. N. Zelonis, N. Mangine, K. Sunderland, S. A. Maas, S. Ching, Y. Barak-Corren, D. W. Laurence, **W. Wu**, P. M. Sabin, A. Lasso, M. Gillespie, J. A. Weiss, M. A. Jolley. "Simulation of Self-Expanding Transcatheter Pulmonary Valve Deployment in the Right Ventricular Outflow Tract," 2024 Summer Biomechanics, Bioengineering and Biotransport Conference, Lake Geneva, Wisconsin, US. June 2024. [Poster].
9. P. M. Sabin, D. W. Laurence, **W. Wu**, C. Herz, S. A. Maas, J. A. Weiss, M. A. Jolley. "Evaluation of Transcatheter Edge-to-Edge Repair Clip Selection via an Open-Source Finite Element Simulation Framework," 2024 Summer Biomechanics, Bioengineering and Biotransport Conference, Lake Geneva, Wisconsin, US. June 2024. [Poster].
8. **W. Wu**, M. Daneker, M. A. Jolley, K. T. Turner, L. Lu. "Effective Physics-Informed Machine Learning Strategies for Material Identification," 17th U.S. National Congress on Computational Mechanics, Albuquerque, New Mexico, US. July 2023. [Oral].
7. **W. Wu** and L. Lu. "Machine Learning for Material Designs," MACH 2023, Baltimore, Maryland, US. June 2023. [Oral].
6. **W. Wu** and C. J. Earls. "Towards a Generalized Engineering Theory for Hydrodynamic Slamming Emanating from Partitioned Fluid-Structure Interaction Analysis," 16th U.S. National Congress on Computational Mechanics, Virtual. July 2021. [Oral].
5. **W. Wu** and C. J. Earls. "Tightly Coupled, Partitioned Fluid-Structure Interaction Analysis of a Horizontal Plate Impact onto a Water Free Surface: Computational Framework and Validation," 15th U.S. National Congress on Computational Mechanics, Austin, Texas, US. July 2019. [Oral].
4. **W. Wu** and C. J. Earls. "Open Source, Tightly Coupled, Partitioned Fluid-Structure Interaction Modeling Framework for Naval Applications: The Impact of Slamming Loads on High Speed Watercraft," 13th World Congress on Computational Mechanics, New York City, New York, US. July 2018. [Oral].
3. P. J. Hughes, W. Scott, **W. Wu**, R. J. Kuether, M. S. Allen, and P. Tiso. "Interface Reduction on Hurty / Craig-Bampton Substructures with Frictionless Contact," IMAC Annual Meeting, Orlando, Florida, US. February 2018. [Oral].

2. **W. Wu**, J. W. Kosianka, and C. J. Earls. "Open Source, Tightly Coupled, Partitioned Fluid-Structure Interaction Simulation Capability for High Spatiotemporal Resolution During Study of Wave Impact Loads in High Speed Watercraft," 14th U.S. National Congress on Computational Mechanics, Montreal, Canada. July 2017. [Oral].
1. J. W. Kosianka, **W. Wu**, and C. J. Earls. "Condition Assessment and Prognosis using Fluid-Structure Interaction within a Reduced-Order Model Tracking Inversion Framework," 14th U.S. National Congress on Computational Mechanics, Montreal, Canada. July 2017. [Oral].

WORKSHOP PRESENTATION

1. **W. Wu**. "Determining Heterogeneous Elastic Properties of Biological Tissues using Physics-Informed Neural Networks," the NSF AI Institute for Artificial Intelligence and Fundamental Interactions Summer Workshop at MIT, Boston, Massachusetts. August 2024.

LEADERSHIP EXPERIENCE

Children's Hospital of Philadelphia / University of Pennsylvania

Biomedical Postdoctoral Council at the University of Pennsylvania | *Career and Training Committee* 2022–2023

Cornell University

CEE Graduate Student Association | *Vice President* 2020–2021

Engineering Learning Initiatives program | *TA Development Consultant* 2018–2019

CEE Graduate Student Association | *Treasurer* 2016–2017

Chi Epsilon National Civil Engineering Honor Society | *Treasurer* 2014–2015

American Society of Civil Engineers | 2014 ASCE Upstate NY Region Student Conference Committee 2013–2014

TEACHING EXPERIENCE

Children's Hospital of Philadelphia / University of Pennsylvania

Multi-institutional Placenta Biomechanics Workshop | *Facilitator* Spring 2024

- Participating institutions included Perelman School of Medicine at the University of Pennsylvania and Vanderbilt University

Cornell University

CEE 4740: Introduction to The Behavior of Metal Structures | *Teaching Assistant* Spring 2019

CEE 3720: Intermediate Solid Mechanics | *Guest Lecturer* Summer 2018

CEE 4780/6780: Structural Dynamics and Earthquake Engineering | *Teaching Assistant* Spring 2018

MENTORING EXPERIENCE

University of Pennsylvania

Rishabh Mallela | *Undergrad in Computer Science* 2025–Present

Tricia Lobo | *MS student* 2024–Present

Mohsen Nakhaei | *Research associate scientist* 2024–Present

Iris Wong | *Undergrad in Bioengineering* Fall 2025

Children's Hospital of Philadelphia

Nishan Parvez | *Postdoc fellow* 2025–Present

Jalaj Maheshwari | *Research staff scientist* 2023–2026

Nicolas Mangine | *Research technician* 2023–2025

Christopher N. Zelonis | *Research specialist* 2023–2025

Silvani Amin | *Research assistant* | Now PhD student at Penn 2021–2023

Stephen Ching | *Research assistant* | Now PhD student at Penn 2021–2023

PROFESSIONAL MEMBERSHIPS

American Society of Civil Engineers
Biomedical Engineering Society
U.S. Association for Computational Mechanics
Society for Industrial and Applied Mathematics

PROFESSIONAL SERVICES

Conference Minisymposium Co-organizer

2026 SIAM Conference on the Life Sciences

Manuscript Reviewer

APL Machine Learning; Journal of the Mechanical Behavior of Biomedical Materials; Journal of the Mechanics and Physics of Solids; Journal of the Royal Society Interface; Medical Engineering and Physics.

Abstract Reviewer

Science Slam Competition at the Children's Hospital of Philadelphia (2023); Intersections Science Fellows Symposium (2023); Biomedical Postdoctoral Research Symposium at the University of Pennsylvania (2022).